

BunkVauth: Next-Generation SSO Security

Security Analysis and Comparison with Traditional SSO Systems

1. The Shift to Zero-Trust Authentication

Traditional SSO creates a single point of failure. BunkVauth addresses this by moving from static, one-time authentication to dynamic, continuous verification.

Security Comparison Table

Feature	Traditional SSO	BunkVauth (Improved)
Authentication Type	One-time (Login only)	Continuous (Real-time monitoring)
Credential Type	Passwords / SMS OTP	FIDO2 Biometrics (Unphishable)
Session Security	Binary (Trusted until expiry)	Risk-based (Zero-Trust scoring)
Theft Protection	Vulnerable to Cookie Theft	Protected by Behavioral Signature
User Experience	Interrupted by repeated prompts	Seamless background verification

2. Key Security Pillars

- FIDO2/WebAuthn Integration:** Eliminates passwords entirely, using hardware-backed biometrics that are mathematically impossible to phish.
- Continuous Behavioral Biometrics:** Captures "How you act" rather than just "What you know". This creates a second layer of security that follows the user throughout the session.
- AI Risk Scoring Engine:** Uses Random Forest ML to calculate an anomaly score. This allows the system to detect account takeover in under 5 seconds.

3. Summary for Project Defense

BunkVauth transforms the SSO model into a proactive security framework. By implementing Hybrid Continuous Authentication, it effectively mitigates Session Hijacking and Credential Theft, making it a robust solution for enterprise-grade identity management.